

Michal J. Wallace

Software engineer · formal methods · Lean 4

Hicksville, NY

CONTACT
michal.wallace@gmail.com
(678) 812-9126

tangentstorm.com
github.com/tangentstorm

SKILLS
Formal methods

Lean 4 (interactive theorem proving), Mathlib, binary decision diagrams, SAT solving

Languages

Python, JavaScript/TypeScript, Rust, K, SQL, Smalltalk, Dart/Flutter

Web & APIs

Vue.js, React, Node.js, HTMX, MCP / LLM tooling, API design

Data

PostgreSQL, MySQL, SQLite, CouchDB

Practices

TDD, Agile, scalable architecture, developer platforms

CODE
laptop
conway-knots
turyn
bex
jprez
b4
pileUp

ELSEWHERE
@tangentstorm (YouTube)
@tangentstream
ArrayCast guest (Nov 2022)
@tangentstorm on X

SUMMARY
Software engineer with deep, hands-on experience in formal methods — interactive theorem proving in Lean 4 across two book-scale formalization projects — plus 20+ years building production web, backend, and AI-integrated systems. Strengths in Lean 4, Python, JavaScript/TypeScript, Rust, and array languages (K/J/APL).

FORMAL METHODS
Method: autoformalization — AI agents draft the Lean 4 proofs, the proof assistant checks them, and failures feed back into the next attempt. I orchestrate the loop and verify the results; the proofs below were not written by hand.

laptop Lean 4 formalization of Eric Hehner's "A Practical Theory of Programming". Full-book blueprint rendered with Verso; all six optional sections plus the \$7.1.5 residual proved, "lake build" green. (repo, site)

conway-knots Lean 4 formalization of Conway's classification of rational tangles. 362-item blueprint at 100% coverage of planned scope. A verified knot enumeration program from the formalization is planned future work. (repo, site)

bex Rust crate for boolean expressions: expression trees and binary decision diagrams, the core data structure of symbolic model checking. SAT-solver and combinatorial-search background. (repo)

turyn Search engine for Turyn-type sequences (four $\{+1,-1\}$ sequences whose combined aperiodic autocorrelations vanish), building blocks for Hadamard matrices via the base-sequence $\rightarrow$ T-sequence $\rightarrow$ Goethals–Seidel pipeline; long-term goal TT(56), yielding a Hadamard matrix of order 668. SAT-driven pipeline (persistent CDCL solver, MDD boundary enumeration, parallel workers), including an AI-created SAT solver. (repo)

EXPERIENCE
Founder
Minneron LLC · Hicksville, NY (Remote)
Jan 2024 – Present

- Formalizing mathematics in Lean 4 (see Formal Methods above). (minavo.app, make.minneron.com)
- Building voiceover and animation software for video creators.
- AI-related consulting projects alongside product work.

Senior Director, Software Development
Symphony AI · Remote
Jun 2025 – May 2026

- Added natural-language support via LLMs to the company's data analysis platform.
- Built an MCP server so LLMs can integrate with that platform.
- Built a proxy to share and monitor AI API key usage.

Senior Software Development Engineer
1010data · NYC Metro (Hybrid)
May 2016 – Jan 2024

- Full-stack web and API development with K, Python, JavaScript, and in-house languages.
- Created a user-extensible API for building applications on the platform.

Software Engineer (Contract)
Balfour · Dallas, TX
Jan 2016 – May 2016

- Desktop publishing for school yearbooks in Flask (Python) and AngularJS.

Small Business Owner / Freelance Developer
Cornerhost (Sabren Enterprises, Inc.) · Atlanta Metro
Nov 2001 – May 2016

- Ran a web hosting company end-to-end (control panel, billing, monitoring) mostly in Python.
- Side contracts across Python, Perl, PHP, and Java (WordPress/Drupal, Android/App Engine, radio ad tooling).

PREVIOUSLY
Developer/Manager at StomperNet (Drupal/WordPress, Flash video library; introduced VCS and bug tracking). Java/XP developer at Media Ocean (TV ad negotiation + EJB $\leftrightarrow$ DHTML messaging). Lead developer/analyst at Abel Solutions (Fortune 500 / NGO / state govt intranet apps). Trainer/developer at Nuesoft; QA analyst at First Command. Self-taught since middle school; AmeriCorps literacy tutor; UT Dallas Art & Performance + Cognitive Science.